

2.29 Numerical Fluid Mechanics

Spring 2015 – Lecture 10

REVIEW Lecture 9:

- **End of (Linear) Algebraic Systems**
 - Gradient Methods
 - Krylov Subspace Methods
 - Preconditioning of $Ax=b$
- **FINITE DIFFERENCES**
 - Classification of Partial Differential Equations (PDEs) and examples with finite difference discretizations
 - Parabolic PDEs
 - Elliptic PDEs
 - Hyperbolic PDEs

FINITE DIFFERENCES - Outline

- Classification of Partial Differential Equations (PDEs) and examples with finite difference discretizations
 - Parabolic PDEs, Elliptic PDEs and Hyperbolic PDEs
- **Error Types and Discretization Properties**
 - Consistency, Truncation error, Error equation, Stability, Convergence
- **Finite Differences based on Taylor Series Expansions**
 - Higher Order Accuracy Differences, with Example
 - Taylor Tables or Method of Undetermined Coefficients
- **Polynomial approximations**
 - Newton's formulas
 - Lagrange polynomial and un-equally spaced differences
 - Hermite Polynomials and Compact/Pade's Difference schemes
 - Equally spaced differences
 - Richardson extrapolation (or uniformly reduced spacing)
 - Iterative improvements using Roomberg's algorithm

References and Reading Assignments

- Chapter 23 on “Numerical Differentiation” and Chapter 18 on “Interpolation” of “Chapra and Canale, Numerical Methods for Engineers, 2006/2010/2014.”
- Chapter 3 on “Finite Difference Methods” of “J. H. Ferziger and M. Peric, Computational Methods for Fluid Dynamics. Springer, NY, 3rd edition, 2002”
- Chapter 3 on “Finite Difference Approximations” of “H. Lomax, T. H. Pulliam, D.W. Zingg, *Fundamentals of Computational Fluid Dynamics (Scientific Computation)*. Springer, 2003”

Partial Differential Equations

Hyperbolic PDE: $B^2 - 4AC > 0$

Examples:

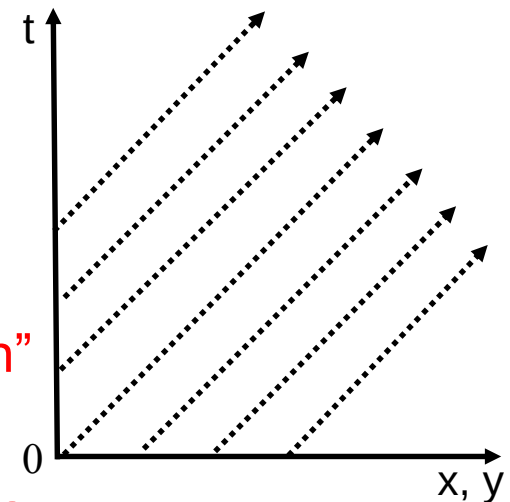
(1) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ $\leftarrow$ Wave equation, 2nd order

(2) $\frac{\partial u}{\partial t} \pm c \frac{\partial u}{\partial x} = 0$ $\leftarrow$ Sommerfeld Wave/radiation equation, 1st order

(3) $\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{U} \cdot \nabla) \mathbf{u} = \mathbf{g}$ $\leftarrow$ Unsteady (linearized) inviscid convection (Wave equation first order)

(4) $(\mathbf{U} \cdot \nabla) \mathbf{u} = \mathbf{g}$ $\leftarrow$ Steady (linearized) inviscid convection

- Allows non-smooth solutions
- Information travels along characteristics, e.g.:
 - For (3) above: $\frac{d \mathbf{x}_c}{dt} = \mathbf{U}(\mathbf{x}_c(t))$
 - For (4), along streamlines: $\frac{d \mathbf{x}_c}{ds} = \mathbf{U}$
- Domain of dependence of $\mathbf{u}(\mathbf{x}, T) = \text{“characteristic path”}$
 - e.g., for (3), it is: $\mathbf{x}_c(t)$ for $0 < t < T$
- Finite Differences, Finite Volumes and Finite Elements



Partial Differential Equations

Hyperbolic PDE - Example

Waves on a String

$$\frac{\partial^2 u(x,t)}{\partial t^2} = c^2 \frac{\partial^2 u(x,t)}{\partial x^2} \quad 0 < x < L, \quad 0 < t < \infty$$

Initial Conditions

$$u(x, 0) = f(x), \quad 0 \leq x \leq L$$

$$u_t(x, 0) = g(x), \quad 0 < x < L$$

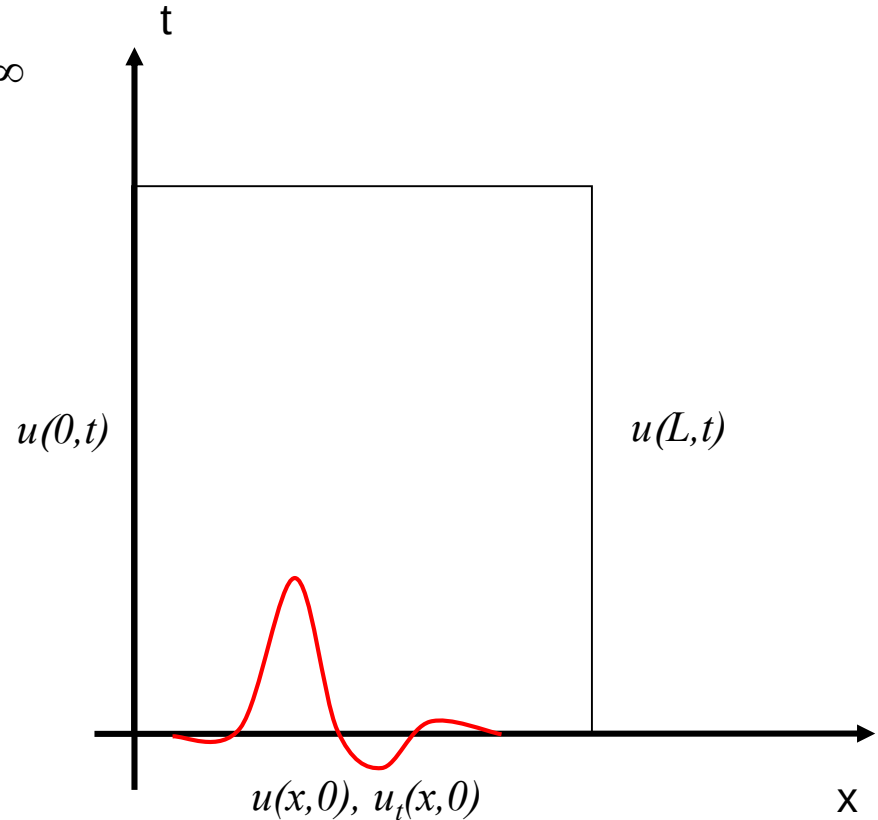
Boundary Conditions

$$u(0, t) = 0, \quad 0 < t < \infty$$

$$u(L, t) = 0, \quad 0 < t < \infty$$

Wave Solutions

$$u = \begin{cases} F(x - ct) & \text{Forward propagating wave} \\ G(x + ct) & \text{Backward propagating wave} \end{cases}$$



Typically Initial Value Problems in Time, Boundary Value Problems in Space

Time-Marching Solutions:

Implicit schemes generally stable

Explicit sometimes stable under certain conditions

Partial Differential Equations

Hyperbolic PDE - Example

Wave Equation

$$\frac{\partial^2 u(x,t)}{\partial t^2} = c^2 \frac{\partial^2 u(x,t)}{\partial x^2} \quad 0 < x < L, \quad 0 < t < \infty$$

Discretization: $h = L/n$

$$k = T/m$$

$$x_i = (i-1)h, \quad i = 2, \dots, n-1$$

$$t_j = (j-1)k, \quad j = 1, \dots, m$$

Finite Difference Representations (centered)

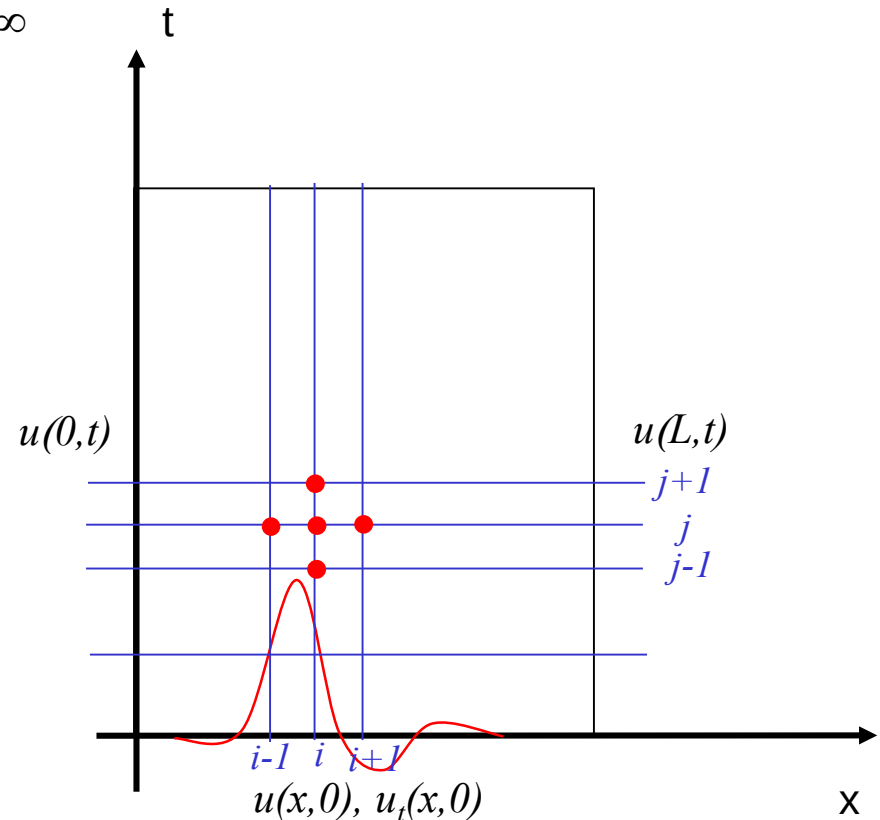
$$u_{tt}(x, t) = \frac{u(x_i, t_{j-1}) - 2u(x_i, t_j) + u(x_i, t_{j+1}))}{k^2} + O(k^2)$$

$$u_{xx}(x, t) = \frac{u(x_{i-1}, t_j) - 2u(x_i, t_j) + u(x_{i+1}, t_j))}{h^2} + O(h^2)$$

$$u_{i,j} = u(x_i, t_j)$$

Finite Difference Representations

$$\frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{k^2} = c^2 \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2}$$



Partial Differential Equations

Hyperbolic PDE - Example

Introduce Dimensionless Wave Speed $C = \frac{ck}{h}$

Explicit Finite Difference Scheme

$$u_{i,j-1} - 2u_{i,j} + u_{i,j+1} = C^2(u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

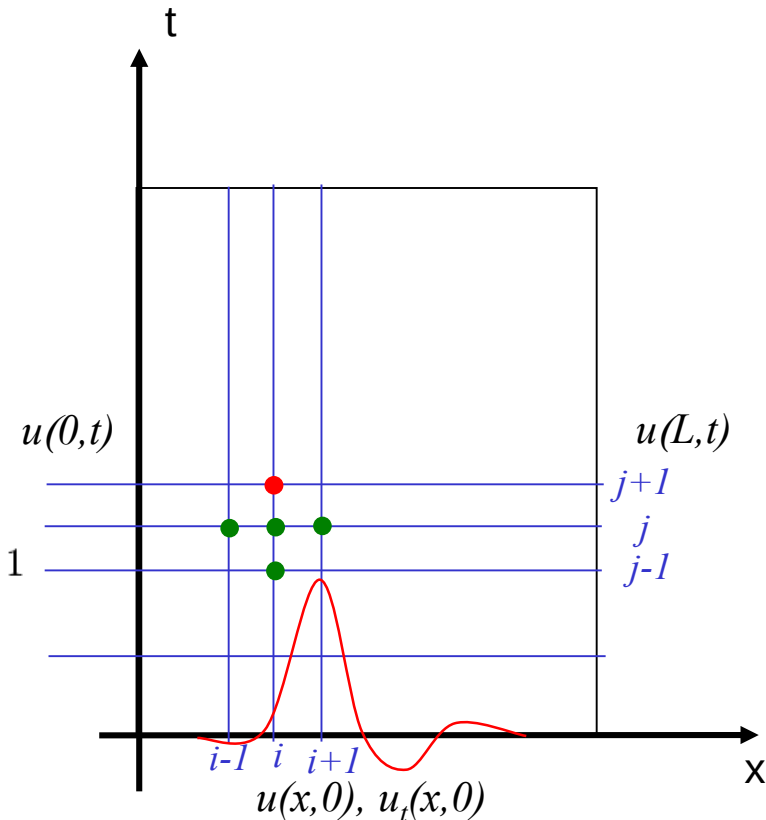
$$u_{i,j+1} = (2 - 2C^2)u_{i,j} + C^2(u_{i+1,j} + u_{i-1,j}) - u_{i,j-1} \quad i = 2, \dots, n-1$$

Stability Requirement: $C = \frac{ck}{h} < 1$

$$C = \frac{c \Delta t}{\Delta x} < 1 \quad \text{Courant-Friedrichs-Lewy condition (CFL condition)}$$

Physical wave speed must be smaller than the largest numerical wave speed, or,
Time-step must be less than the time for the wave to travel to adjacent grid points:

$$c < \frac{\Delta x}{\Delta t} \quad \text{or} \quad \Delta t < \frac{\Delta x}{c}$$



Error Types and Discretization Properties:

Consistency

Consider the differential equation ($\mathcal{L}$ symbolic operator)

$$\mathcal{L}(\phi) = 0$$

and its discretization for any given difference scheme

$$\hat{\mathcal{L}}_{\Delta x}(\hat{\phi}) = 0$$

❖ Consistency (Property of the discretization)

- The discretization of a PDE should asymptote to the PDE itself as the mesh-size/time-step goes to zero, i.e

for all smooth functions ϕ :
$$\left| \mathcal{L}(\phi) - \hat{\mathcal{L}}_{\Delta x}(\phi) \right| \rightarrow 0 \quad \text{when } \Delta x \rightarrow 0$$

(the truncation error vanishes as mesh-size/time-step goes to zero)

Error Types and Discretization Properties:

Truncation error and Error equation

❖ Truncation error

$$\tau_{\Delta x} = \mathcal{L}(\phi) - \hat{\mathcal{L}}_{\Delta x}(\phi)$$

Remember:

ϕ does not satisfy the FD eqn.

- Since $\mathcal{L}(\phi) = 0$, the truncation error is the result of inserting the exact solution in the difference scheme
- If the FD scheme is consistent: $\tau_{\Delta x} = \mathcal{L}(\phi) - \hat{\mathcal{L}}_{\Delta x}(\phi) \rightarrow O(\Delta x^p)$ for $\Delta x \rightarrow 0$
- $p (> 0)$ is the order of accuracy for the FD scheme $\hat{\mathcal{L}}_{\Delta x}$
- Order p indicates how fast the error is **reduced** when the grid is **refined**

❖ Error evolution equation

- From $\hat{\mathcal{L}}_{\Delta x}(\hat{\phi}) = 0$ and $\phi = \hat{\phi} + \varepsilon$ where ε is the discretization error, for linear problems, we have:
$$\tau_{\Delta x} = \mathcal{L}(\phi) - \hat{\mathcal{L}}_{\Delta x}(\hat{\phi} + \varepsilon) = -\hat{\mathcal{L}}_{\Delta x}(\varepsilon)$$
$$\Rightarrow \hat{\mathcal{L}}_{\Delta x}(\varepsilon) = -\tau_{\Delta x}$$
- The truncation error acts as a source for the discretization error, which is convected, diffused, evolved, etc., by the operator $\hat{\mathcal{L}}_{\Delta x}$

Error Types and Discretization Properties: Stability

❖ Stability

- A numerical solution scheme is said to be stable if it does not amplify errors ε that appear in the course of the numerical solution process
- For linear(-ized) problems, since $\hat{\mathcal{L}}_{\Delta x}(\varepsilon) = -\tau_{\Delta x}$, stability implies:

$$\left\| \hat{\mathcal{L}}_{\Delta x}^{-1} \right\| < \text{Const.} \quad \text{with the Const. not a function of } \Delta x$$

- If inverse was not bounded, discretization errors ε would increase with iterations
- In practice, infinite norm $\left\| \hat{\mathcal{L}}_{\Delta x}^{-1} \right\|_{\infty} < \text{Const.}$ is often used.
- However, difficult to assess stability in real cases due to boundary conditions and non-linearities
 - It is common to investigate stability for linear problems, with constant coefficients and without boundary conditions
 - A widely used approach: von Neumann's method (see lectures 12-13)

Error Types and Discretization Properties: Convergence

❖ Convergence

- A numerical scheme is said to be convergent if the solution of the discretized equations tend to the exact solution of the (P)DE as the grid-spacing and time-step go to zero

- Error equation for linear(-ized) systems: $\varepsilon = -\hat{\mathcal{L}}_{\Delta x}^{-1}(\tau_{\Delta x})$

- Error bounds for linear systems:

$$\|\varepsilon\| = \|\hat{\mathcal{L}}_{\Delta x}^{-1}(\tau_{\Delta x})\| \leq \|\hat{\mathcal{L}}_{\Delta x}^{-1}\| \|\tau_{\Delta x}\|$$

For a consistent scheme: $\|\tau_{\Delta x}\| \rightarrow O(\Delta x^p)$ for $\Delta x \rightarrow 0$

$$\text{Hence } \|\varepsilon\| \leq \|\hat{\mathcal{L}}_{\Delta x}^{-1}\| \|\tau_{\Delta x}\| \leq \alpha O(\Delta x^p)$$

Convergence $\leq$ Stability + Consistency (for linear systems)

= Lax Equivalence Theorem (for linear systems)

- For nonlinear equations, numerical experiments are often used
 - e.g., iterate or approximate true solution with computation on successively finer grids, and compute resulting discretization errors and order of convergence

Finite Differences - Basics

- Finite Difference Approximation idea directly borrowed from the definition of a derivative.

$$\phi'(x_i) = \lim_{\Delta x \rightarrow 0} \frac{\phi(x_{i+1}) - \phi(x_i)}{\Delta x}$$

- Geometrical Interpretation

- Quality of approximation improves as stencil points get closer to x_i
- Central difference would be exact if ϕ was a second order polynomial and points were equally spaced

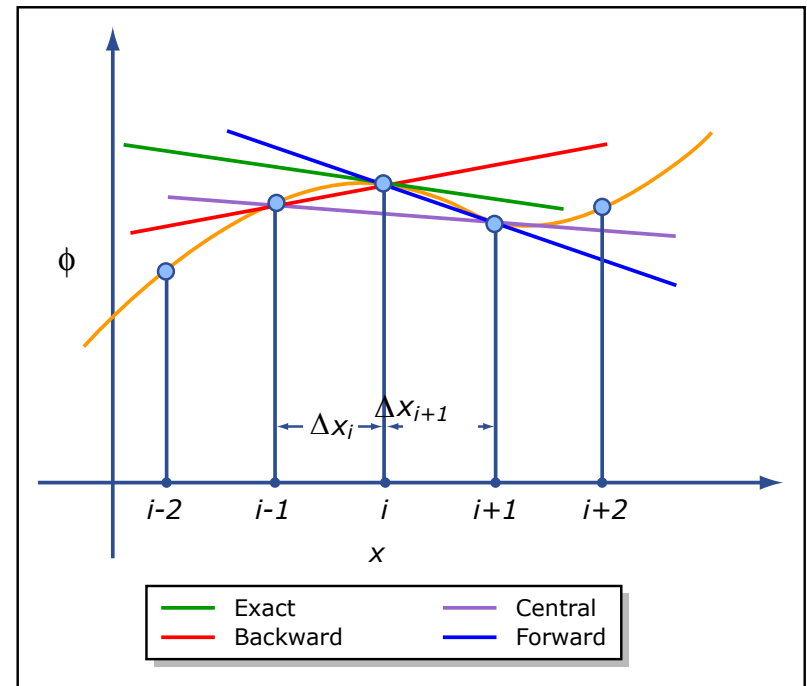


Image by MIT OpenCourseWare.

On the definition of a derivative and its approximations

FINITE DIFFERENCES:

Taylor Series, Higher Order Accuracy

How to obtain differentiation formulas of arbitrary high accuracy?

- 1) First approach: Use Taylor series, keep more higher-order terms than strictly needed and express these higher-order terms as finite-differences themselves

$$f(x_{i+1}) = f(x_i) + \Delta x f'(x_i) + \frac{\Delta x^2}{2!} f''(x_i) + \frac{\Delta x^3}{3!} f'''(x_i) + \dots + \frac{\Delta x^n}{n!} f^n(x_i) + R_n$$

$$R_n = \frac{\Delta x^{n+1}}{(n+1)!} f^{(n+1)}(\xi)$$

- For example, how can we derive the forward finite-difference estimate of the first derivative at x_i with second order accuracy?

$$\left. f(x_{i+1}) = f(x_i) + \Delta x f'(x_i) + \frac{\Delta x^2}{2!} f''(x_i) + O(\Delta x^3) \right\} \longrightarrow f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{\Delta x} - \frac{\Delta x}{2!} \underline{f''(x_i)} + O(\Delta x^2)$$

- If we retain the second-derivative, and estimate it with first-order accuracy, the order of accuracy for the estimate of $f'(x_i)$ will be $p=2$

FINITE DIFFERENCES:

Taylor Series, Higher Order Accuracy Cont'd

Still using $f(x_{i+1}) = f(x_i) + \Delta x f'(x_i) + \frac{\Delta x^2}{2!} f''(x_i) + \frac{\Delta x^3}{3!} f'''(x_i) + \dots + \frac{\Delta x^n}{n!} f^n(x_i) + R_n$

$$R_n = \frac{\Delta x^{n+1}}{n+1!} f^{(n+1)}(\xi)$$

- Estimate the second-derivative with forward finite-differences at first-order accuracy:

$$\left. \begin{aligned} f(x_{i+1}) &= f(x_i) + \Delta x f'(x_i) + \frac{\Delta x^2}{2!} f''(x_i) + O(\Delta x^3) \\ f(x_{i+2}) &= f(x_i) + 2\Delta x f'(x_i) + \frac{4\Delta x^2}{2!} f''(x_i) + O(\Delta x^3) \end{aligned} \right\} \begin{array}{l} * (-2) \\ * (1) \end{array} \Rightarrow f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{\Delta x^2} + O(\Delta x)$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{\Delta x} - \frac{\Delta x}{2!} f''(x_i) + O(\Delta x^2)$$

$$\Rightarrow \underline{f'(x_i)} = \frac{f(x_{i+1}) - f(x_i)}{\Delta x} - \frac{\Delta x}{2!} \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{\Delta x^2} + O(\Delta x^2) = \underline{\underline{\frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2\Delta x} + O(\Delta x^2)}}}$$

Forward finite-divided-difference formulas: two versions are presented for each derivative. The latter version incorporates more terms of the Taylor series expansion and is, consequently, more accurate.

Figure 23.1
Chapra and
Canale

Forward Differences

First Derivative

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h}$$

Error

$O(h)$

$$f'(x_i) = \frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h}$$

$O(h^2)$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2}$$

$O(h)$

$$f''(x_i) = \frac{-f(x_{i+3}) + 4f(x_{i+2}) - 5f(x_{i+1}) + 2f(x_i)}{h^2}$$

$O(h^2)$

Third Derivative

$$f'''(x_i) = \frac{f(x_{i+3}) - 3f(x_{i+2}) + 3f(x_{i+1}) - f(x_i)}{h^3}$$

$O(h)$

$$f'''(x_i) = \frac{-3f(x_{i+4}) + 14f(x_{i+3}) - 24f(x_{i+2}) + 18f(x_{i+1}) - 5f(x_i)}{2h^3}$$

$O(h^2)$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_{i+4}) - 4f(x_{i+3}) + 6f(x_{i+2}) - 4f(x_{i+1}) + f(x_i)}{h^4}$$

$O(h)$

$$f^{(4)}(x_i) = \frac{-2f(x_{i+5}) + 11f(x_{i+4}) - 24f(x_{i+3}) + 26f(x_{i+2}) - 14f(x_{i+1}) + 3f(x_i)}{h^4}$$

$O(h^2)$

Backward Differences

FIGURE 23.2
Backward finite-divided-difference formulas: two versions are presented for each derivative. The latter version incorporates more terms of the Taylor series expansion and is, consequently, more accurate.

First Derivative

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h}$$

$$f'(x_i) = \frac{3f(x_i) - 4f(x_{i-1}) + f(x_{i-2})}{2h}$$

Second Derivative

$$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2}))}{h^2}$$

$$f''(x_i) = \frac{2f(x_i) - 5f(x_{i-1}) + 4f(x_{i-2}) - f(x_{i-3}))}{h^2}$$

Third Derivative

$$f'''(x_i) = \frac{f(x_i) - 3f(x_{i-1}) + 3f(x_{i-2}) - f(x_{i-3}))}{h^3}$$

$$f'''(x_i) = \frac{5f(x_i) - 18f(x_{i-1}) + 24f(x_{i-2}) - 14f(x_{i-3}) + 3f(x_{i-4}))}{2h^3}$$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_i) - 4f(x_{i-1}) + 6f(x_{i-2}) - 4f(x_{i-3}) + f(x_{i-4}))}{h^4}$$

$$f^{(4)}(x_i) = \frac{3f(x_i) - 14f(x_{i-1}) + 26f(x_{i-2}) - 24f(x_{i-3}) + 11f(x_{i-4}) - 2f(x_{i-5}))}{h^4}$$

Error

$$O(h)$$

$$O(h^2)$$

$$O(h)$$

$$O(h^2)$$

$$O(h)$$

$$O(h^2)$$

$$O(h)$$

$$O(h^2)$$

3.3

Centered finite-divided-difference formulas: two versions are presented for each derivative. The latter version incorporates more terms of the Taylor series expansion and is, consequently, more accurate.

Centered Differences

First Derivative

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}$$

Error

$O(h^2)$

$$f'(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2}))}{12h}$$

$O(h^4)$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2}$$

$O(h^2)$

$$f''(x_i) = \frac{-f(x_{i+2}) + 16f(x_{i+1}) - 30f(x_i) + 16f(x_{i-1}) - f(x_{i-2}))}{12h^2}$$

$O(h^4)$

Third Derivative

$$f'''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + 2f(x_{i-1}) - f(x_{i-2}))}{2h^3}$$

$O(h^2)$

$$f'''(x_i) = \frac{-f(x_{i+3}) + 8f(x_{i+2}) - 13f(x_{i+1}) + 13f(x_{i-1}) - 8f(x_{i-2}) + f(x_{i-3}))}{8h^3}$$

$O(h^4)$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_{i+2}) - 4f(x_{i+1}) + 6f(x_i) - 4f(x_{i-1}) + f(x_{i-2}))}{h^4}$$

$O(h^2)$

$$f^{(4)}(x_i) = \frac{-f(x_{i+3}) + 12f(x_{i+2}) + 39f(x_{i+1}) + 56f(x_i) - 39f(x_{i-1}) + 12f(x_{i-2}) + f(x_{i-3}))}{6h^4}$$

$O(h^4)$

FINITE DIFFERENCES

Taylor Series, Higher Order Accuracy: **EXAMPLE**

Problem: Estimate 1st derivative of $f = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$ at $x=0.5$, with a grid cell size of $h=0.25$ and using successively higher order schemes. How does the solution improve?

```
L11_FD.m
%Define the function
f=@(x) -0.1*x^4 - 0.15*x^3-0.5*x^2-0.25*x +1.2;
%Define Step size
h=0.25;
%Set point at which to evaluate the derivative
x = 0.5;
%% Using forward difference
%First order:
df=(f(x+h)-f(x)) / h;
fprintf('\n\n First order Forward difference: %g, with
error:%g%% \n',df,abs(100*(df+0.9125)/0.9125))
%Second order:
df=(-f(x+2*h)+4*f(x+h)-3*f(x)) / (2*h);
fprintf('Second order Forward difference: %g, with
error:%g%% \n',df,abs(100*(df+0.9125)/0.9125))
%% Backwards difference
%First order:
df=(-f(x-h)+f(x)) / (h);
fprintf('First order Backwards difference: %g, with
error:%g%% \n',df,abs(100*(df+0.9125)/0.9125))
%Second order:
df=(f(x-2*h)-4*f(x-h)+3*f(x)) / (2*h);
fprintf('Second order Backwards difference: %g, with
error:%g%% \n',df,abs(100*(df+0.9125)/0.9125))
```

```
%% Central difference
%Second order:
df=(f(x+h)-f(x-h)) / (2*h);
fprintf('Second order Central difference: %g, with
error:%g%% \n',df,abs(100*(df+0.9125)/0.9125))
%Fourth order:
df=(-f(x+2*h)+8*f(x+h)-8*f(x-h)+f(x-2*h)) /
(12*h);
fprintf('Fourth order Central difference: %g, with
error:%g%% \n',df,abs(100*(df+0.9125)/0.9125))
```

Output

First order Forward difference: -1.15469, with error:26.5411%
Second order Forward difference: -0.859375, with error:5.82192%
First order Backwards difference: -0.714063, with error:21.7466%
Second order Backwards difference: -0.878125, with error:3.76712%
Second order Central difference: -0.934375, with error:2.39726%
Fourth order Central difference: -0.9125, with error:2.43337e-14%

Why is the 4th order “exact”?

FINITE DIFFERENCES:

Taylor Series, Higher Order Accuracy

Summary

- 1st Approach:
 - Incorporate more higher-order terms of the Taylor series expansion than strictly needed and express them as finite differences themselves
 - e.g. for finite difference of m^{th} derivative at order of accuracy p , express the $m+1^{\text{th}}$, $m+2^{\text{th}}$, $m+p-1^{\text{th}}$ derivatives at an order of accuracy $p-1, \dots, 2, 1$.
 - General approximation:
$$\left(\frac{\partial^m u}{\partial x^m} \right)_j - \sum_{i=-r}^s a_i u_{j+i} = \tau_{\Delta x}$$
 - Can be used for forward, backward, skewed or central differences
 - Can be computer automated
 - Independent of coordinate system and extends to multi-dimensional finite differences (each coordinate is often treated separately)
- Remember: order p of approximation indicates how fast the error is **reduced** when the grid is **refined** (not necessarily the magnitude of the error)

FINITE DIFFERENCES:

Interpolation Formulas for Higher Order Accuracy

2nd approach: Generalize Taylor series using interpolation formulas

- Fit the unknown function solution of the (P)DE to an interpolation curve and differentiate the resulting curve. For example:

- Fit a parabola to “ f data” at points x_{i-1}, x_i, x_{i+1} ($\Delta x_i = x_i - x_{i-1}$), then differentiate to obtain:

$$f'(x_i) = \frac{f(x_{i+1})(\Delta x_i)^2 - f(x_{i-1})(\Delta x_{i+1})^2 + f(x_i)[(\Delta x_{i+1})^2 - (\Delta x_i)^2]}{\Delta x_{i+1} \Delta x_i (\Delta x_i + \Delta x_{i+1})}$$

- This is a 2nd order approximation (parabola approx. is of order 3)
- For uniform spacing, reduces to centered difference seen before
- In general, approximation of first derivative has a truncation error of the order of the polynomial (here 2)
- All types of polynomials or numerical differentiation methods can be used to derive such interpolations formulas
 - Polynomial fitting, Method of undetermined coefficients, Newton's interpolating polynomials, Lagrangian and Hermite Polynomials, etc

FINITE DIFFERENCES Higher Order Accuracy: Taylor Tables or Method of Undetermined Coefficients

Taylor Tables: Convenient way of forming linear combinations of Taylor Series on a term-by-term basis

Table 3.1. Taylor table for centered 3-point Lagrangian approximation to a second derivative

What we are looking for, in 1st column:

$$\left(\frac{\partial^2 u}{\partial x^2}\right)_j - \frac{1}{\Delta x^2}(a u_{j-1} + b u_j + c u_{j+1}) = ?$$

		u_j	$\Delta x \left(\frac{\partial u}{\partial x}\right)_j$	$\Delta x^2 \left(\frac{\partial^2 u}{\partial x^2}\right)_j$	$\Delta x^3 \left(\frac{\partial^3 u}{\partial x^3}\right)_j$	$\Delta x^4 \left(\frac{\partial^4 u}{\partial x^4}\right)_j$
Taylor series at:	$\Delta x^2 \left(\frac{\partial^2 u}{\partial x^2}\right)_j$			1		
j-1	$-a \cdot u_{j-1}$	$-a$	$-a \cdot (-1) \cdot \frac{1}{1}$	$-a \cdot (-1)^2 \cdot \frac{1}{2}$	$-a \cdot (-1)^3 \cdot \frac{1}{6}$	$-a \cdot (-1)^4 \cdot \frac{1}{24}$
j	$-b \cdot u_j$	$-b$				
j+1	$-c \cdot u_{j+1}$	$-c$	$-c \cdot (1) \cdot \frac{1}{1}$	$-c \cdot (1)^2 \cdot \frac{1}{2}$	$-c \cdot (1)^3 \cdot \frac{1}{6}$	$-c \cdot (1)^4 \cdot \frac{1}{24}$

Sum each column starting from left, force the sums to zero and so choose a, b, c, etc

FINITE DIFFERENCES

Higher Order Accuracy: Taylor Tables Cont'd

Table 3.1. Taylor table for centered 3-point Lagrangian approximation to a second derivative

$$\left(\frac{\partial^2 u}{\partial x^2}\right)_j - \frac{1}{\Delta x^2}(a u_{j-1} + b u_j + c u_{j+1}) = ?$$

Sum each column starting from left and force the sums to be zero by proper choice of a, b, c, etc:

$$\begin{bmatrix} -1 & -1 & -1 \\ 1 & 0 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix} \Rightarrow [a \quad b \quad c] = [1 \quad -2 \quad 1] \quad \text{= Familiar 3-point central difference}$$

Truncation error is first column in the table that does not vanish, here fifth column of table:

$$\tau_{\Delta x} = \frac{1}{\Delta x^2} \left[\frac{-a}{24} + \frac{-c}{24} \right] \Delta x^4 \left(\frac{\partial^4 u}{\partial x^4} \right)_j = -\frac{\Delta x^2}{12} \left(\frac{\partial^4 u}{\partial x^4} \right)_j$$

FINITE DIFFERENCES

Higher Order Accuracy: Taylor Tables Cont'd

Table 3.2. Taylor table for backward 3-point Lagrangian approximation to a first derivative

$$\left(\frac{\partial u}{\partial x} \right)_j - \frac{1}{\Delta x} (a_2 u_{j-2} + a_1 u_{j-1} + b u_j) = ?$$

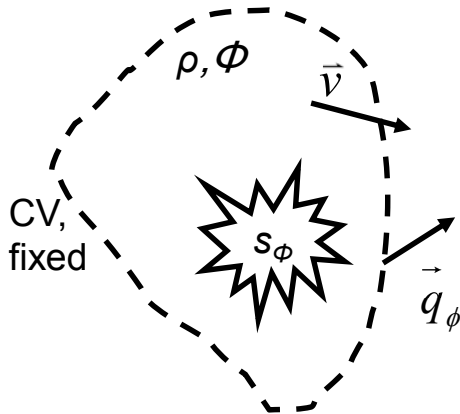
	u_j	$\Delta x \left(\frac{\partial u}{\partial x} \right)_j$	$\Delta x^2 \left(\frac{\partial^2 u}{\partial x^2} \right)_j$	$\Delta x^3 \left(\frac{\partial^3 u}{\partial x^3} \right)_j$	$\Delta x^4 \left(\frac{\partial^4 u}{\partial x^4} \right)_j$
$\Delta x \left(\frac{\partial u}{\partial x} \right)_j$		1			
$-a_2 \cdot u_{j-2}$	$-a_2$	$-a_2 \cdot (-2) \cdot \frac{1}{1}$	$-a_2 \cdot (-2)^2 \cdot \frac{1}{2}$	$-a_2 \cdot (-2)^3 \cdot \frac{1}{6}$	$-a_2 \cdot (-2)^4 \cdot \frac{1}{24}$
$-a_1 \cdot u_{j-1}$	$-a_1$	$-a_1 \cdot (-1) \cdot \frac{1}{1}$	$-a_1 \cdot (-1)^2 \cdot \frac{1}{2}$	$-a_1 \cdot (-1)^3 \cdot \frac{1}{6}$	$-a_1 \cdot (-1)^4 \cdot \frac{1}{24}$
$-b \cdot u_j$	$-b$				

$$\Rightarrow [a_2 \quad a_1 \quad b] = [1 \quad -4 \quad 3]/2 \quad \text{and} \quad \tau_{\Delta x} = \frac{1}{\Delta x} \left[\frac{8a_2}{6} + \frac{a_1}{6} \right] \Delta x^3 \left(\frac{\partial^3 u}{\partial x^3} \right)_j = \frac{\Delta x^2}{3} \left(\frac{\partial^3 u}{\partial x^3} \right)_j$$

(as before)

Integral Conservation Law for a scalar ϕ

$$\left\{ \frac{d}{dt} \int_{CM} \rho \phi dV = \right\} \frac{d}{dt} \int_{CV_{\text{fixed}}} \rho \phi dV + \underbrace{\int_{CS} \rho \phi (\vec{v} \cdot \vec{n}) dA}_{\substack{\text{Advective fluxes} \\ (\text{Adv. \& diff. fluxes} \Rightarrow \text{convection})}} = \underbrace{- \int_{CS} \vec{q}_\phi \cdot \vec{n} dA}_{\substack{\text{Other transports (diffusion, etc)}}} + \underbrace{\sum \int_{CV_{\text{fixed}}} s_\phi dV}_{\substack{\text{Sum of sources and} \\ \text{sinks terms (reactions, etc)}}$$



Applying the Gauss Theorem, for any arbitrary CV gives:

$$\frac{\partial \rho \phi}{\partial t} + \nabla \cdot (\rho \phi \vec{v}) = -\nabla \cdot \vec{q}_\phi + s_\phi$$

For a common diffusive flux model (Fick's law, Fourier's law):

$$\vec{q}_\phi = -k \nabla \phi$$

Conservative form
of the PDE

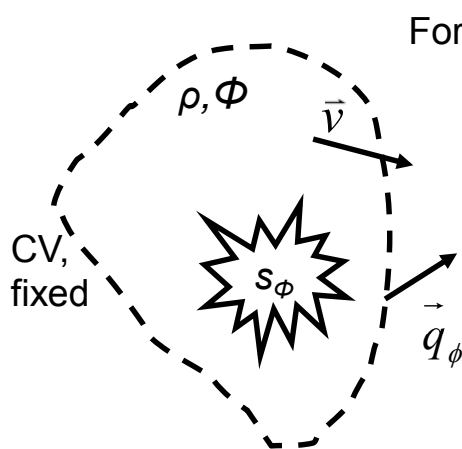
$$\frac{\partial \rho \phi}{\partial t} + \nabla \cdot (\rho \phi \vec{v}) = \nabla \cdot (k \nabla \phi) + s_\phi$$

Strong-Conservative form of the Navier-Stokes Equations ($\phi \Rightarrow \mathbf{v}$)

Cons. of Momentum:
$$\frac{d}{dt} \int_{CV} \rho \vec{v} dV + \int_{CS} \rho \vec{v} (\vec{v} \cdot \vec{n}) dA = \underbrace{\int_{CS} -p \vec{n} dA + \int_{CS} \vec{\tau} \cdot \vec{n} dA + \int_{CV} \rho \vec{g} dV}_{=\sum \vec{F}}$$

Applying the Gauss Theorem gives:

$$= \int_{CV} (-\nabla p + \nabla \cdot \vec{\tau} + \rho \vec{g}) dV$$



For any arbitrary CV gives:

$$\frac{\partial \rho \vec{v}}{\partial t} + \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla p + \nabla \cdot \vec{\tau} + \rho \vec{g} \quad \text{Cauchy Mom. Eqn.}$$

With Newtonian fluid + incompressible + constant μ :

$$\begin{aligned} \text{Momentum: } & \frac{\partial \rho \vec{v}}{\partial t} + \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla p + \mu \nabla^2 \vec{v} + \rho \vec{g} \\ \text{Mass: } & \nabla \cdot \vec{v} = 0 \end{aligned}$$

Equations are said to be in “strong conservative form” if all terms have the form of the divergence of a vector or a tensor. For the i^{th} Cartesian component, in the general Newtonian fluid case:

$$\text{With Newtonian fluid only: } \frac{\partial \rho v_i}{\partial t} + \nabla \cdot (\rho v_i \vec{v}) = \nabla \cdot \left(-p \vec{e}_i + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \vec{e}_j - \frac{2}{3} \mu \frac{\partial u_j}{\partial x_j} \vec{e}_i + \rho g_i x_i \vec{e}_i \right)$$

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Spring 2015

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